Jeremy Ku-Benjet

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SUMMARY

Jeremy is research masters student interested in compilers, programing langauges, and EDA tools. He's currently working with Adrian Sampson on on accelerating analytical placement using FPGAs and creating build tools to simplify writing design flows. Previously, he worked on backend engineering for a course management system. Jeremy has course work in ASIC design, accelerator design with FPGAs, microcontrollers, compilers, programming languages, distributed systems, and theoretical linear algebra.

SKILLS AND TECH

- **Programming/RTL Languages:** Rust, C++, Python, Java, OCaml, Mercury (similar to Prolog), Verilog, RISCV and x86-64 Assembly, Bash.
- **Technologies:** LLVM, Quartus, Git, Unix Shell Scripting, CMake, Make, Ninja, Docker/Podman.
- **Human Language:** English (Native).

EXPERIENCE

fud2 | Researcher with Adrian Sampson **Jun 2024 — Present**

- Developed an extensible compiler driver
- Extend the Rhai scripting language to integrate fud2 with new tools
- Wrote an algorithm to automatically synthesize tool flows from simple cli commands
- To be presented at LATTE'26, a workshop about hardware acceleration and programming languages co-located with ASPLOS

CMSX | Backend Developer **Feb 2024 — Jan 2026**

- Worked on a course management system used by thousands of students and professors at Cornell University
- Lead design and implementation of a rubrics feature
- Wrote backend logic for grading rubrics
- Audit a transaction based database system for errors when rolling back invalid database queries
- Add a calendar feature allowing students to download deadlines using ical calendar format

Teaching Assistant | Cornell University **Jan 2023 — Present**

- Improve assignments and course infrastructure, design exam questions, staff office hours, teach lab sections, and grade for courses on Computer Architecture, Compilers, Systems Programming, Object Oriented Programming, Functional Programming, and Game Design.
- Develop and test a tool to manage and run students' devlopment containers
- Debug extensions to the LibGDX game library

PROJECTS

- **laft: Logic Analyzer:** Designed a SUMP compatible logic analyzer sampling at up to 100MHz. Built in C using an Raspberry PI Pico 2 and a custom PCB. Wrote a protocol separate from SUMP for communication over USB. Implemented a GUI for displaying waveforms using the new protocol.
- **Apathia:** Programming lead on an online head to head mobile strategy game, written in C++.
- **Seas the Throne:** Programming lead on an action adventure bullet hell, written in Java.

EDUCATION

Cornell University **Aug 2022 — May 2025**

BS - Computer Science, GPA: 3.8

Cornell University **Aug 2025 — Present**

MS - Computer Science, GPA: 4.0